

Section 2 - 3D Essentials Layout & User Settings



Lecture 2 | Layout & Workspace

WORKSPACE

This is the CLO workspace. The interface is similar to other softwares as there is a menu bar along the top of the screen. Each menu has a drop down of options. Under file you will find all your saving, importing, and exporting options.

- Edit will give you editing options that would otherwise be accessed with quick commands or by Right-Clicking.
- The 3D Garment, 2D Pattern, Sewing and Materials tabs are all alternative ways of accessing tools and options that can also be found in your toolbars or by Right-Clicking in the 2D and 3D backgrounds.
- The Editor Menu is new to version 7. Two options which were previously in the Modes Drop down menu are now here; Bill of Materials and Colorways. Plus there is a new Editor called Parametric Pattern Editor
- The Avatar menu gives you options for editing your Avatar. Some of these options can also be found in the 3D toolbar.
- Render is going to take you to the internal Render engine or allow you to render via CLO-set.
- Display allows you to hide or show different things in your 2D or 3D windows. Most of these options are also located in the vertical toggle bars in each window. I'll go over those in full at the end. You also can hide or show toolbars located in your 3D or 2D window.
- Preferences allows you to customize settings within the program like your gizmo, simulation properties, fit maps, or camera properties. Many of these options are also available by Right-Clicking in the 2D or 3D backgrounds.
- Settings will allow you to change your language, adjust user settings, access a plug-in, or log out of the software.
- Help is going to give you options for accessing information to help guide you in using the program.

In the far-right corner of the menu bar, you will see your login username, the Share to CLOSET icon and Drop Down with all the CLO Modes.

LIBRARY

Starting In the top left is the Library. There are three tabs, a STAR, C Cloud and the CLOSET Connect logo. The C Cloud will link to your CLOSET account. The Connect Logo is linked to CLOSET Connect and the items you have purchased will be available in the "My Items" folder.

The star tab contains all the default assets that come with the program: things like Avatars, Garments, Hardware and Trims etc. In addition to the default assets, you also can link folders from your computer into your library so that you can access things quickly and easily. To link a folder, click the plus sign in the top right-hand corner of the library. Then navigate to the folder that you want to add. You don't need to click into the folder, simply select it. Now my folder has been added to the bottom of the list of folders in my library.

To the left of the add button is the download button. This button may have a blue dot beside it when you first download or update the software. You will also see blue dots next to the folders as well. The blue dot indicates that there are default library assets available that you have not yet downloaded. Click the download button to update your library to the latest assets of CLO.

New to version 7, the CLO Library now has a tree structure that allows you to see the folders inside of folders. When you start CLO, the folders are always collapsed. Next to any folder which contains other folders is an arrow that when clicked shows the contents. You can continue to open folders where arrows appear and Double-Click to see the contents in the explorer below. Note that "Stage" does not have an arrow; that is because there are no folders within. Click the arrows again to collapse the folders.

To open a folder in the library just double click it. The folder's contents show up in this area below; this is like the sidebar in a normal window in your computer. Double click the sub folders as much as you need. The folder icon at the top with the 2 dots beside it will take you back one folder. To open something from your library, just double click on it or right-click and you will get a set of options depending on what that file is. For things like fabrics and print images, you can simply click and drag and drop them where you need them to go in CLO; this is one of the benefits of adding your own folders from your desktop to have easy access.

You can view all library folders as icons or lists by clicking this button. When you hover your mouse over the edge of the library you get an icon with 2 arrows that allows you to expand or collapse the library. You can fully collapse the library at the top right by clicking the arrow in the library bar. To get it back, click Library on the black bar on the left. The shortcut for this is the letter "Y" on your keyboard. "Y" also brings the library back.

If you make any changes to the folders linked to your library, click the refresh button here. This will update all the folders in your library to match those on your computer.

The search bar on the left allows you to search by name for a folder or file within the folder that you're already in. This drop-down arrow allows you to sort your folders by category and arrange them in ascending or descending order. Be sure to clear this search box when finished.

Lastly in the top right-hand corner is the reset button. This will take you back to the default library and delete any folders that have been added.

On the left hand side there is a toolbar that shows the history tab and modular configurator.

The history tab will show you all of the actions you've taken in this file since opening. You can select an action to move back to the step in the file. If you'd like to save the project at a certain

point but continuing working you can use 3D state. At the top right corner you can select the 3D state button. This saves a copy of the file within the project that is accessible only here. The 3D states will save with your project file and you can open at any point. You can continue working within the file without changing the saved 3D state.

By Right-Clicking you can delete a 3D State or rename it. You can also rename it by just clicking in the text. You can save multiple 3D states throughout your file. Close the history by clicking on the arrow in the top right-hand corner.

Also in the left side bar is the modular configurator. This is a block library that allows you to mix and match different parts of a garment and create a block to work from.

You'll notice after clicking into it, even if you don't use it, you'll get these grey shapes in your 2D background. Right-click in the 2D background and select "Remove Modular Relationship" to get rid of these.

3D & 2D WINDOWS

To the left of center is your 3D window. This is where your avatar and your 3D garment will be. The toolbar at the Left are all your 3D tools. To the right of center is your 2D pattern window. Your 2D pattern and 3D garment are one and the same, each respective window just allows you to view it in different states. The toolbar at the right of the 2D window are all your 2D tools.

If you hover the mouse over the center, you get the icon with the select / move tool icon that allows you to adjust the size of the windows. You also have the ability in the top right-hand corner to undock the windows. This allows you to move one to a different monitor or you can rearrange them left to right if you prefer the 2D on the left.

TOGGLE BARS

In the upper left corner of the 3D and 2D windows you have vertical toggle menus that allow you to hide and show different attributes within those windows. In 3D you have attributes related to the garment, Garment Details, the Avatar, fabric views, Fit maps, and different options for viewing the surface texture of your avatar.

By hovering over the double black bars within the tool bars, you can undock that set of tools and rearrange them as you wish. If you want your entire toolbar to the top or bottom of the 3D & 2D windows, you can change that in User Settings. Please watch the next video in this series to understand user settings.

In 2D you have garment detail, pattern options, information and measurements, and different options for viewing the texture of your pattern pieces. If you hover over any of the icons, it will tell you exactly what it does and if there is a hot key associated with that tool. And if anything, ever looks like it's missing or is showing something, you don't want to see, always make sure you check these menus first.

MATERIALS

The area in the top right-hand corner is the Object Browser. This is like your Bill of Materials. It houses a scene tab, your fabrics, graphics, buttons, buttonholes, topstitchings, puckering, grading and point of measure; each of these are represented by an icon. If you hover over the icon in the tab, the full tool name will appear. If you click on something in the Object Browser, its properties are listed below in the Property Editor.

The Scene tab will be familiar to you if you have worked in other 3D software. This allows you to navigate to specific parts of your 3D project file that you may otherwise have trouble finding including smaller properties like trims or other properties that you may want to edit. You are also able to remove items from this menu by holding delete and

As mentioned before, the Property Editor displays all the properties of what you select and allows you to edit them. This is true of things within the object browser, as well as anything that you click on in the 2D or 3D window. Most things in the program will have editable properties.

EXTRAS

At the bottom right-hand corner, there are buttons to see only your 3D, only your 2D, or both windows at the same time. The button on the far right is your reset button. This is going to reset the layout of the entire workspace to the default arrangement. This is helpful if you find you're missing a tool or have moved around windows.

At the bottom left-hand corner, you will see the current version of CLO that you are using.

User Settings

GRAPHIC OPTIONS

Let's have a look at the User Settings. Go to the Settings Menu→ User Settings. This window will appear.

I am going to cover the most common settings that might be changed. Should you need want information about any of these options, have a look at the PDF document for this lesson which is included with the resources. [You can also go to our CLO Support Forum.](#)

The first section, Graphic Options, refers to the visual settings in your 3D Window.

"Use VBO's" (or Vertex Buffer Objects) correlates to the rendering speed. It is a faster way of rendering, but if your computer does not have a graphics card that can handle it, you may experience issues while rendering. This could include black flickering in the background of your 3D window while working or other issues. If this is the case, you should uncheck this box. This is one of the settings that will ask you to restart CLO. This just means that the change won't go

into effect until you close and re-open CLO, but you can do that at any time after making the change. You don't have to do it immediately.

Antialiasing is common with gaming. The number corresponds to speed and the smoothing of curves and edges. If your screen doesn't have a high enough resolution to process the graphics, you get jagged edges and antialiasing will smooth it out. The default is x16. There should be no need to adjust this unless you are seeing an issue with the graphics in your 3D window.

NORMAL BLENDING

Normal blending takes a collection of the normal maps used in a style and changes how the maps are blended together. Normal Blending Method allows you to choose from a drop down list of different options. You can see the blending result in the 3D window and the UV editor.

- Alpha blending the default in the software, which simulates the normal blending effect with the alpha value.
- Partial Derivative blends the height map using partial derivatives of the normal map.
- Whiteout uses a technique optimized to reduce the amount of computation and it is a way to preserve detail while maintaining the tendency of a relatively flat normal.
- UDN - Compared to Whiteout and Reoriented, the detail of the normal map is slightly reduced but the optimized calculation method is useful for low-end platforms.
- Reoriented - A method that preserves the intensity of all mixed normal maps while maintaining the trend of the flat normal map and the details of the complex normal map well. This is the most accurate and slow method by rotating the normal of the detail to match the normal of the base.
- Unity is the same method as reoriented but the error in certain situations (when two mixed normals are orthogonal) is abandoned and the performance is improved.

VIEW CONTROLS

View Controls refers to your mouse navigation set-up and your point of view in the 3D Window. The default on a PC is a Regular Mouse. If your mouse does not have a scroll wheel, the default zoom will be the Left Mouse Button and Alt. If your mouse has a scroll wheel, your scroll wheel will be your zoom by default.

On a Mac the default will be set to Magic Mouse. The other settings allow you to match your navigation settings to other 3D programs.

Tablet is the setting used for a Wacom Tablet or a Laptop Trackpad. CLO is not yet compatible with a touch screen. User Preset at the top will pop up if any of the drop-down options are customized. There are also settings for Maya, 3DS Max and other input devices.

Axis Constraint Key refers to the key that is held when you want to lock yourself into 45 and 90 degree angles. This might be locking your axis of rotation in 3D or used when pattern editing,

among other things. By default, this key is Shift. You can choose different keys from the drop-down menu.

Smart panning allows you to pan while doing other things without interrupting that action, for example when sewing in 2D. I find this nearly essential for CLO workflow, so I do not recommend unchecking this box.

Auto Focus means that if you click on something, it becomes your central point of rotation in 3D. If this is unchecked, the center is always the same- 0-0-0 where your Avatar loads. The option below it will revert to center as the point of rotation once everything is de-selected. If unchecked, you will need to hit the 2 key to bring your point of rotation back to center. The video on Navigation and Selection covers this in greater detail.

These are the default settings for your Viewpoint in the 3D window. If changed you will see your 3D "eye level" moves up and down, or tilts forward and backward. This number will change the distance from your avatar and garment when using the default viewpoints.

SHORTCUTS

The shortcuts menu allows you to customize keyboard shortcuts. To do this, select the shortcut and you can simply delete it or if you click on the shortcut itself, you just need to type in the key that you want to be the shortcut. If you want it to be a combination of keys, you need to press them both at the same time. The reset button will reset the shortcut to the default. Please know CLO will not allow you to set shortcuts that are already in use, so you will need to remove it where it exists by default, before adding it somewhere else.

Down here at the bottom you will notice a different shortcut for the QWE gizmo when adding pins. That's because the W shortcut has its own function with that gizmo. For more on this, skip ahead to the 3D settings section of this video.

USER INTERFACE

Unit

- The User Interface tab firstly allows you to customize your units of measure within the software.
- Here you can check this box to use ligne as the units for button width. If you don't use ligne button width will default to your chosen unit of measure.
- Here you can select the units for thread thickness.
- Currency can be changed from the drop down list and is used in the BOM mode.

Mode & Toolbar

- Default mode allows you to choose which mode you'd like to open the file in. Next time you open the file it will open in the mode selected from the drop down list.
- Here you can change the placement of your tool bars in the 3D and 2D windows. These are the default settings but can be changed based on preference.
- Group Tools nests similar tools under one main tool. If you have a small screen keeping this off may help.

- This box determines whether the learning links pop up when you hover over a tool in the tool bar. These links take you to our website and show either a video or written description of the tool. Here you can change the font used within the program. Using these is a good way to further your learning of CLO.

Appearance

- You CAN change which font is used in the CLO interface. If this is changed and you see abnormalities, be sure to change back to Tahoma.
- You can change the visual appearance of selected objects here. In these lessons we use the default options, though Thickness and Line Length Size may be a bit larger than your defaults.
- Automatic scale will optimize your visual display in most cases. If your screen needs some more adjusting you can try from the drop down menu and select QHD, FHD or 4K. These options WILL NOT be available on MAC. You will need to adjust your Display Resolution in the System Preferences.

3D

Mesh

Now let's look at the 3D tab. Here you can adjust whether the default mesh is triangular or quad, meaning rectangular. In 3D software, everything is made of mesh. In CLO, by default, we use a triangular mesh, as it is optimal for getting the soft drape of fabrics and the curves of the human body. You can change the type of mesh on an individual pattern piece in the property editor but if you change it here, it will change your default for creating any patterns.

Arrangement

Under arrangement you can change the default arrangement setting of your pattern pieces from curved to flat. By default, when using arrangement points, the pattern pieces will curve around the avatar. You can also change this on individual pieces in the property editor if you do not need to change it for everything. You can see more on this in the Avatar & Arrangement Video. You can also change the Arrangement Point Size as well.

Sync fold arrangement makes sure that when something is folded using the fold arrangement tool in 3D, the fold angle is adjusted accordingly in the property editor.

By default, the gizmo is a unified gizmo that allows you to select, move and rotate all at once. If you select divided gizmo, you will need to hold the Q key to select, the W key to move and the E key to rotate. Remember if you select this gizmo, the keyboard shortcut for adding an individual pin will have to change.

2D

In the 2D menu, you can adjust the default particle distance, which is set to 20mm.

Checking this box means that when you cut 2 patterns apart in the 2D window, they will automatically shift to put a small space between them. If unchecked, they will stay touching where they are cut.

These two options mean that your pattern notches will visually get bigger as you zoom out, to see them from far away. Or they will stay the same size and be to scale, no matter how you zoom.

OTHER

Now I'm in Other Settings. Under Modular Configurator, this setting when checked on will allow you to change seamlessly between blocks when the naming convention has been set up to do so, like going from a single to double breasted jacket using the same sleeves. When unchecked, switching from one block to another will require you to start over.

This option below will maintain your fabrics between blocks if you have already set those.

These settings are for connecting directly from CLO to a plotter and they'll depend on the plotter you are using.

This option allows you to set the download location of DMIX Cloud files.

This area shows the applications you have added to the Open Texture option. You can add and remove here as well.

DEFAULT FILES & PLUG-IN (CLO API)

Default files show the file location where things are automatically saved or pulled from. You can adjust the file path by clicking the folder icon and you can reset back to the default folder by clicking the arrow.

PROJECT; where the default project is saved. You can customize a project and save here to have your own project layout. You also have the option to have a Multiview thumbnail. If checked, you can choose from a default layout or Custom File (MVS).

AVATAR; If you are using custom avatars that need permission files, you can add those permissions here.

MATERIALS; Specifies where and how various Fabrics & Textures are saved.

SIMULATION PROPERTY; default location of SMP file

AUTO SAVE; Here you can adjust the time interval that CLO autosaves. By default, it is set to save every 10 minutes. You can also reference the file path in case you didn't get the pop-up to open the autosaved file in CLO and you want to search for it on your computer.

CUSTOM COLOR PALETTE; you can create and save your own custom palette and save in this location.

GARMENT INFORMATION; JSON file locations.

If you have or create your own plug-in for CLO, you can access it here through the plug-in menu by clicking the plus sign. To delete a plug-in you can click the trashcan icon.

CLO API

This shows the PlugIns you have installed in CLO.

CLOSET

Lastly the CLO-set tab allows you to link your CLO-set URL automatically from CLO. This used to allow you to pre-login to your CLOSET account so that when uploading or rendering directly out of CLO, it would go straight to your account. This created issues with company CLOSET accounts and now takes you directly to the login screen instead. If your company has its own CLOSET URL, you can enter that here instead of the CLO-SET.com URL.

[Intro to CLOSET](#)

Lecture 3 | Navigation & Selection

Hi everyone, this lesson covers Navigation and Selection. You can follow along with any CLO Project file

INTRO

- Review how to link this folder to your library.
 - Select the + (add) button and navigate to the folder & Select folder
- Double click to open or Right-Click to add the project file into the workspace.
 - Right-Click, you have the option to open or add a project file.
 - Combine Files
 - Default add at the 0,0,0 point which represents the X, Y and Z axis.
- The 2D & 3D windows are linked
- Avatar in the 3D window; 2D window by a silhouette outline of the avatar.
 - Silhouette is proportional to the avatar in the 3D window; used as a guide when positioning or drafting patterns in both windows

NAVIGATION

Let's begin navigating in our 3D window.

- Commands based on a regular 3 button mouse; recommended, because it allows you to navigate entirely with one hand.
 - Easiest if you're a beginner.

- User Settings to set up your device if anything I show here doesn't seem to be working for you.
 - PC, your user settings will be set to a Regular 3-button mouse.
 - Mac, you will have the option to use a Magic Mouse.
 - Select any device from the drop down menu in user settings and see the appropriate keys to follow along.

ZOOM

- Middle scroll wheel to zoom in and out.
- Both your 3D & 2D windows.
- Magic Mouse or a trackpad; 2 fingers and drag up and down or however you currently scroll on that device.

ROTATION

- Rotating is only applicable in the 3D window.
- Right-Click, hold and drag
- Hold shift to lock your axis
 - X or Y axis
- Magic Mouse or trackpad; the same
 - Depending on what version or generation you have of your laptop or mouse, you may need to work with the settings of this on your computer

PAN

- Hold down your scroll wheel button and then move around your screen.
 - Same in 2D or 3D.
- Magic Mouse or a Trackpad; Option key on a Mac or the Alt key on a PC, while Left-Clicking and moving.

RESET YOUR VIEW

- Get back to default window views
- Right-Click in 3D background,
- Viewpoints of your 3D window.
- Front, $\frac{3}{4}$ Left... hot keys associated with them.
- Imagine the avatar is standing on the 5 key of a number keypad
- 2 is probably the most useful; front and center; perfectly straight on
- 2D window, to reset your view, you can Right-Click in the 2D background; "Zoom Extends All". OR Shift-zero - right parenthesis
 - Clip your window to everything that you have in it.
- focus zoom = F key
 - in 3D center the pattern piece in window

- in 2D fills the window with the selected pattern piece

Selection

-
- Main selection tools; Select/Move tool (Hotkey Q) in the 3D Window, Transform Pattern tool (Hotkey A) in the 2D window.
 - “Home Tools”
 - Selected on a tool highlights blue
 - On both of my selection tools.
 - Only 3D Tools, only 2D Tools
 - Crossover tools select in both windows [sewing]
 - Select = Left-Click.
 - Pattern Highlights yellow in both windows
 - Select in 3D, little blue dot in the 2D window. This is called a picking point.
 - Geographic location of selection in 3 D window
 - Select in each window, that same pattern will highlight in both windows.
 - In 3D it highlights in 2D and vice versa.
 - 2D window patterns are highlighted in a brighter blue outline; symmetric copy.
 - No highlights, no symmetric copy.
 - Front and back, so they’re not linked, left and right are linked.
 - Edit one, the other side is going to edit
 - Move the pieces, both pieces will need to be selected.
 - Multi-select, hold down shift and select the pattern pieces.
 - Going back and forth between 2D and 3D.
 - Select something that’s already selected to de-select it
 - Multi-select in 2D is to marquee over things. Y
 - You will need to marquee over the piece entirely.
 - partial selection will not select at all
 - When you multi-select in 3D, you will see that you do get a picking point for every pattern
 - Use CMD or CTRL Z to edit undo any pieces you accidentally move
 - CMD or CTRL Z to undo back to the beginning of the project
 - Note History palette
 - Select all is CMD or CTRL A; Mac or PC.
 - Option for selection = Right-Click on something in the 2D window, at the very top menu you get the option “Select all with same property.”
 - Same layer, same particle distance, same fabric.
 - More about Particle Distance and layering in future lessons.
 - Most useful; “select all with same fabric”.
 - Delete anything; backspace key or the delete key.
 - Same with your avatar.
 - Clear your entire screen with that.

GIZMO

- Select any pattern piece in either window, you will get this tool in the 3D window. This is called a "gizmo"
- All 3D software uses a gizmo
- Allow you to move things in 3D when simulation is off, so when things are just solid pieces like this.
 - We'll learn about the gizmo in the lesson about Arrangement.

Thank you for watching and we will see you in the next tutorial